

Measuring the Cost of Living

Mbak 9515: Economics

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Review

- GDP is most common measure of economic activity and the standard of living in the economy
 - ▶ [Twitter](#)
- Three ways to calculate GDP: *production*, *expenditure*, *income* method
 - ▶ the market value of all final goods and services produced for a marketplace during a period of time, within a country's borders
 - ▶ $Y = C + I + G + NX$
 - ▶ value-add method
- Real GDP vs. Nominal GDP

Review

- Nominal GDP: Value of final goods and services evaluated at *current year prices*
 - ▶ It measures the current dollar value of production
- Real GDP: Value of final goods and services evaluated at *base year prices*
 - ▶ Value of dollars exchanged if prices did not change from the prices in an arbitrary (base) year
 - ▶ It measures the actual physical volume of production

Review

Real GDP

	Pizza		Burger	
	Quantity	Price	Quantity	Price
2013	10	<i>\$10</i>	15	<i>\$5</i>
2017	20	\$12	30	\$6

- Nominal GDP in 2013: $10 \times \$10 + 15 \times \$5 = \$175$
- Nominal GDP in 2017: $20 \times \$12 + 30 \times \$6 = \$420$

- Real GDP in 2013: $10 \times \$10 + 15 \times \$5 = \$175$
- Real GDP in 2017: $20 \times \$10 + 30 \times \$5 = \$350$

Readings

- Chapter 6. Measuring the Cost of Living
 - ▶ Consumer price index (CPI) and the inflation rate
 - ▶ Biases in the CPI
 - ▶ Correcting Economic Variables for the Effects of Inflation

Stable Prices

- Goal 2: Low and Stable Prices
 - ▶ maintaining the *real* value of the currency is another important criteria for a well-performed economy
 - ▶ **Inflation**: Prices are rising
 - ▶ **Deflation**: Prices are falling
 - ▶ How much do consumers care about all prices?
- What is the ideal level of inflation/deflation?

Consumer Price Index

- Consumer price index (CPI)
 - ▶ the basic tool to measure the price level in the economy
 - ▶ for any period, a measure of the cost in that period of a standard basket of goods and services *relative to* the cost of the same basket of goods and services in a fixed year
- e.g.) Suppose 2010 is the base year

	2010		2015	
	Price	Quantity	Price	Quantity
Rent	\$750	1	\$945	1
Burger	\$2	60	\$2.5	60
Movie	\$7	10	\$8	10

- ▶ Total expenditure in 2010: $\$750 \times 1 + \$2 \times 60 + \$7 \times 10 = \940
 - ▶ Total expenditure in 2015: $\$945 \times 1 + \$2.5 \times 60 + \$8 \times 10 = \$1,175$
- If the family want to consume the same basket of goods and services in 2015 as in the base year, the spend **\$235** more
 - ▶ in other words, the family spend *25% ($=\$235/\940)* more

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Consumer Price Index

- Bureau of Labor Statistics (BLS) visits thousands of stores and conduct numerous interviews and collect data every month

$$\text{CPI} = \frac{\text{Price of basket of goods and services in current year}}{\text{Price of basket of goods and services in base year}}$$

- ▶ In the previous numerical example,

$$\text{CPI in 2015} = \frac{\$1,175}{\$940} \times 100 = 125$$

- ▶ The cost of living in the year 2015 is *25% higher* than it was in 2010
- ▶ Note that the base-year CPI is always 100

Consumer Price Index

	Pizzas		Corn dog	
	Quantity	Price	Quantity	Price
2013	10	\$10	15	\$6
2017	20	\$12	30	\$5

- CPI in 2017? (the base year is 2013)

- ▶ Step 1) Calculate total expenditure in the base year

$$10 \times \$10 + 15 \times \$6 = \$190$$

- ▶ Step 2) Calculate total expenditure in year 2017 with **the quantities in the base year**

$$10 \times \$12 + 15 \times \$5 = \$195$$

- ▶ Step 3) Divide the total expenditure in year 2017 with the total expenditure in the base year

$$\frac{\$195}{\$190} \times 100 = 102.6$$

- ▶ The CPI gives a measure of **the average level** of prices relative to prices in the base year

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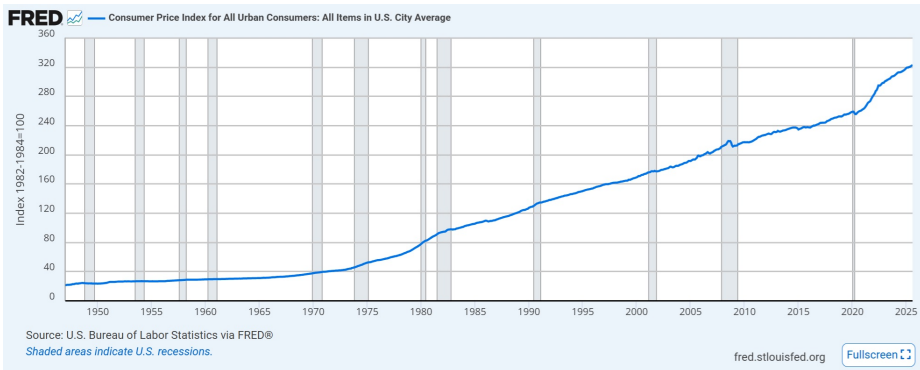
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Consumer Price Index



Inflation

- A measure of how fast the average price level is changing in over time
 - ▶ In the previous numerical example, CPI in year 2017 is 102.6
 - ▶ What is CPI in *the base year*?
 - ▶ **100!**

- The inflation rate between the base year and year 2017

$$\begin{aligned}\text{Inflation rate} &= \frac{102.6 - 100}{100} \times 100 \\ &= 2.6\%\end{aligned}$$

Inflation

Year	CPI
1972	41.8
1973	44.4
1974	49.3

- The inflation rate between year 1972 and 1973: $\frac{44.4-41.8}{41.8} \times 100 = 6.2\%$
- The inflation rate between year 1973 and 1974: $\frac{49.3-44.4}{44.4} \times 100 = 11\%$

Year	CPI
1929	17.1
1930	16.7
1931	15.2

- The inflation rate between year 1929 and 1930:
 $\frac{16.7-17.1}{17.1} \times 100 = -2.3\%$
- The inflation rate between year 1930 and 1931:
 $\frac{15.2-16.7}{16.7} \times 100 = -8.9\%$

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Inflation



Shortcomings of the CPI

• Unmeasured quality change

- ▶ Difficult to separate improvement in quality from increase in price, say in cars or computers
- ▶ iPhone with 20% more memory, computational speed, and data storage capacity than last year's model
- ▶ its price is 20% higher $\Rightarrow$ there's no problem!
- ▶ But CPI simply increases by 20% $\Rightarrow$ *overstate inflation*
- ▶ If we have overstate inflation, living standards would be *underestimated*

• Introduction of new goods

- ▶ The basket of goods changes only every 10 years
- ▶ There is a delay to including new goods like the first effective AIDS drugs

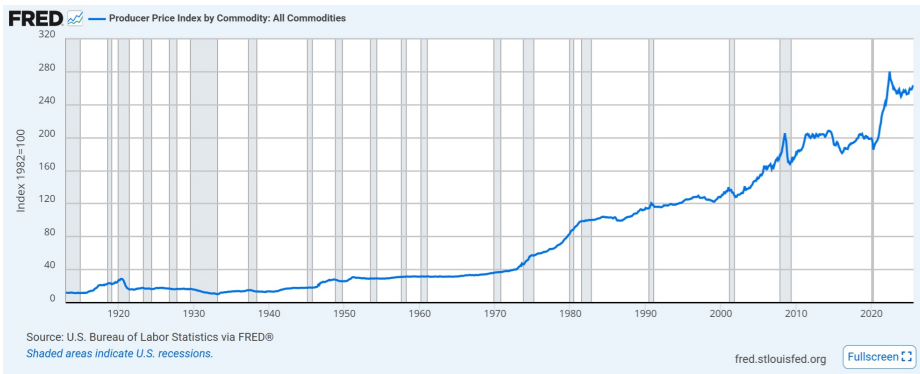
• Substitution bias

- ▶ Consumers may change their purchasing habits away from goods that have increased in price
- ▶ coffee vs. tea

Other Price Indexes

Producer price index (PPI)

- A family of indexes that measures the average change in selling prices received by *domestic producers* of goods and services over time



Other Price Indexes

GDP deflator

- a measure of the level of prices of *all* new, domestically produced, final goods and services in an economy

$$\text{GDP deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$$

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$$2013 : \frac{\$175}{\$175} \times 100 = 100$$

$$2017 : \frac{\$420}{\$350} \times 100 = 120$$

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Other Price Indexes

GDP deflator



Correcting Economic Variables

- Dollar figures from different times
 - ▶ e.g.) income: \$40,000 in 2010 and \$44,000 in 2015
 - ▶ is this economically better off in the year 2015 than in 2010?
- Maybe not! because of price change! What matters is *purchasing power*
 - ▶ We need to calculate incomes in *real terms*

How?

$$\text{Amount in today's dollars} = \text{Amount in year T dollars} \times \frac{\text{Price level today}}{\text{Price level in year T}}$$

Correcting Economic Variables

Who earns more?

- Babe Ruth: \$80,000 in 1931
- Clayton Kershaw: \$33,800,000 in 2017
- We cannot directly compare the *nominal* salaries $\Rightarrow$ We need to know the CPI in year 1930 and year 2017
- CPI in 2015 is 237

Year	Nominal income	CPI	income in 2015
1931	\$80,000	16.7	<i>\$1.2 m</i>
2017	\$33.8 m	245	<i>\$32.7 m</i>

- Kershaw's salary is 27 times more than Ruth's salary
- Incomes are normalized in 2015 dollars

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The Costs of Inflation

- Why do people worry about inflation?
 - ▶ **Price level:** a measure of the overall level of prices at a particular point in time as measured by a price index such as the CPI
 - ▶ **Relative price:** the price of a specific good or service *in comparison to* the prices of other goods and services

The True Costs of Inflation

● Noise in the price system

- ▶ Increasing prices should *signal* increased demand
- ▶ but a high inflation environment interferes with the signal (noise)

● Distortions of the tax system

- ▶ Many taxes are indexed; income tax
- ▶ higher effective tax rates caused by 'bracket creep' in a progressive, non-indexed tax system

● Shoe leather costs

- ▶ The costs associated with holding less cash on hand and more in the bank so as to offset the reduction of purchasing power with interest

● Unexpected redistributions of wealth

Fisher

- ▶ Inflation redistributes rather than destroys purchasing power

● Interference with long-term planning

- ▶ Long-run business and personal decisions become more difficult to make with high and erratic inflation

Hyperinflation

- A situation in which the inflation rate is extremely high
 - ▶ Inflation rates in the range of 500 to 1,000%
 - ▶ Israel (400%), Venezuela (more than 1,000%)
- Germany in 1923
 - ▶ 102,000,000%
 - ▶ workers were paid twice each day
- Hyperinflation greatly magnifies the costs of inflation

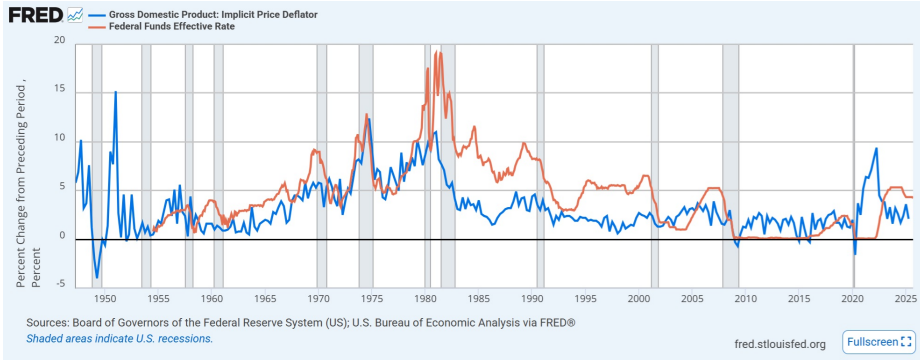
Fisher Equation

- **The nominal interest rate** is the rate of interest specified in the loan contract
- **The real interest rate** is the difference between *the nominal interest rate* and *the inflation rate*
 - ▶ It measures the reward to lenders, in terms of purchasing power, paid by borrowers for lenders giving up spending over the contract period

$$i = r + \pi$$

- ▶ where i is the nominal interest rate, r is the real interest rate, and π is the current inflation rate
- Fisher effect: the tendency for nominal interest rates to be *high* when inflation is *high* and *low* when inflation is *low*

Fisher Effect



Goback